

NEELDRIK

Poora System — Sab Kuch, Aasan Bhasha Me

Satellite se samundar me tel (oil spill) pakadne wala system

SIH26143 · Team **HackSmiths**

Smart India Hackathon 2026

Ye document isliye banaya hai taki aap kisi ko bhi — teacher, judge, dost — ye system samjha sako, bina technical background ke. Har number actual code se check kiya gaya hai.

Isme Kya Milega

1. **Project hai kya** — ek page me poori kahani
2. **Kitne model hain aur kaise bane**
3. **Har model kya use kar raha hai**
4. **Aur wahi kyu use kar rahe hain**
5. **Tech stack** — frontend, backend, poori list
6. **Kaunsi technology kahan lagi** aur kaam ka flow
7. **Aage kya kar sakte hain**
8. **Kya change karne se code me kya badlega**
9. **Result** — aur kya nahi bolna
10. **Judge ke sawaal** — ready jawab

1. Project Hai Kya

Ek line me

Satellite ki radar photo dekhkar batana ki **samundar me tel gira hai ya nahi** — aur agar gira hai to **kahan se aaya hoga**.

Pehle physics samjho — ye base hai

Hawa chalne se samundar par bahut choti-choti lehrein banti hain. Satellite ka radar in lehron se takrata hai aur wapas aata hai. Isliye photo me **samundar safed / bright** dikhta hai.

Ab jab tel girta hai, wo paani par ek patli parat (film) bana leta hai. Ye parat un choti lehron ko **dabaa deti hai** — paani wahan chikna ho jaata hai. Radar chikni jagah se takrata nahi, isliye wapas nahi aata. Isliye photo me **tel kaala** dikhta hai.

Lekin asli problem yahan hai.

Sirf kaala hona kaafi nahi hai. Aur bhi bahut cheezein kaali dikhti hain:

- Hawa ruk jaye (bahut kam wind) → paani chikna → kaala
- Samundar me algae (kai) ho → kaala
- Barish ka cell ho → kaala
- Jahaz ke peeche ka wake → kaala

Inko **"look-alike"** kehte hain — dikhte tel jaise hain, par tel hain nahi.

Isliye poore project ka asli kaam tel dhoondhna nahi hai. Asli kaam hai — kaale dhabbon me se sach me tel kaunsa hai, ye pehchaanna. Baaki sab isi ek problem ke around bana hai.

System ko aise samjho — 5 logon ki team

Socho ek hospital hai jahan X-ray dekhi jaati hai:

Kaun	Kya karta hai	System me iska naam
Reception	"Ye X-ray hai bhi ya selfie bhej di?" Galat cheez ho to andar bhejta hi nahi	Domain Gate
Lab technician	X-ray ko saaf karta hai, dhundhlapan hatata hai, sahi size me laata hai	Preprocessing
Junior doctor (yahi AI hai)	Photo ki har ek jagah par nishaan lagata hai — ye normal, ye tel, ye shak wala	U-Net (Neural Network)
Senior doctor	Junior ki report + mausam dekhkar final faisla deta hai. Confuse ho to "aur jaanch karo" bolta hai	Decision Layer
Physics wala banda	"Tel yahan mila hai — hawa aur paani ka bahaav dekhte hue, 12 ghante pehle ye kahan tha?"	Drift Model

Sabse important baat: in 5 me se **AI sirf ek hai** — junior doctor. Baaki sab maths aur rules hain.

Aur ye galti se nahi hua, jaan-bhoojh kar hai. Tel ka behna hawa aur current se **exactly calculate** ho jaata hai. Jo cheez maths se pakki nikal rahi ho, usko AI se andaaza lagwana bewakoofi hai.

2. Kitne Model Hain Aur Kaise Bane Sawaal 1

#	Model	Kis type ka hai	Kaise bana
1	Domain Gate	Rules (AI nahi)	Simple statistics — kitna grayscale hai, radar wala noise hai ya nahi. Haath se likhe rules.
2	U-Net Ye hi AI hai	Deep Learning (CNN)	Architecture khud likha, backprop ka maths khud likha, phir pehle nakli data par, phir asli satellite data par train kiya.
3	Decision Layer	Calibrated rules	Rules to humne banaye, par unke numbers (thresholds) computer ne data dekhkar khud dhoonde.
4	Drift Model	Physics + Maths	Oceanography ke standard equations code me likhe. Training nahi hui — iski zaroorat hi nahi.
5	Data Generator	Physics simulator	Nakli SAR photos banane wala program — training ke liye.

U-Net kaise bana — step by step

- 1 Architecture design** — "U" shape ka network. Pehle photo ko chhota karte hain (taki "kya hai" samajh aaye), phir wapas bada karte hain (taki "kahan hai" bata sake). Beech me shortcut connections hain taki kinaare (boundaries) saaf rahein.
- 2 Maths khud likha** — Conv2D, BatchNorm, ReLU, MaxPool, Adam optimizer — ye sab NumPy me haath se likhe. Koi ready-made library nahi. Phir **numerical gradient check** se verify kiya ki maths sahi hai (error 0.0002 se kam).
- 3 Phase 1 — nakli data se training.** Shuru me asli labelled data nahi tha. Isliye physics se nakli SAR photos banayi: samundar ka speckle, tel ka dark+smooth patch, look-alike ka dark-par-textured patch, jahaz, zameen. Model ne basic samajh yahan se paayi.
- 4 Phase 2 — asli satellite data se training.** Zenodo se **450 asli Sentinel-1 photos** mile (Trujillo-Acatitla 2024 dataset) — 150 tel wali, 150 look-alike wali, 150 saaf. Har photo 2048×2048 pixel ki thi, aur saath me mask tha jo batata tha tel exactly kahan hai.
- 5 Data ko tukdon me kaata** — badi photo ko 5,268 chote patches me kaata, har patch 128×128 ka. Model ko itni badi photo ek saath nahi di ja sakti.
- 6 Training chalayi** — 40 baar poora data dikhaya (40 epochs), 22,440 steps, laptop CPU

par 45 minute. PyTorch use kiya.

- 7 Thresholds calibrate kiye** — training ke baad computer ne khud dhoonda ki "kitna tel ho tab spill bolein".

Data conversion me 4 bug mile the — agar ye na pakadte to training chup-chaap kharab ho jaati aur pata bhi na chalta:

1. **Folder ka naam** hi `..._and_ground_truth` tha, aur code poore path me "ground_truth" dhoondh raha tha → **har photo ko mask samajh liya**. Result: 0 pairs.
2. **VV/VH ulta tha** — radar do tarah se signal bhejta hai. Code galat wala use kar raha tha. Ab har photo me khud measure karke sahi wala chunta hai.
3. **Zero padding** — jahan satellite ki coverage nahi thi wahan value exactly 0 thi, jo kaali nahi **sabse tez safed** hai. Ek photo to aadhe se zyada khaali thi. Aise 2,370 patches hata diye.
4. **Tukde galat chun rahe the** — pehle jo tukde aa rahe the wo 99% tel hote the. Usse model ko pata hi nahi chalta ki "tel *khatam* kahan hota hai". Ab wo tukde chunte hain jo tel ke **kinaare** par hain.

Ek zaroori baat — cheating kaise roki

Ek hi satellite photo ke saare tukdon me same samundar, same lighting, same noise hota hai.

Agar ek hi photo ke kuch tukde **padhai** me chale jaayein aur kuch **exam** me, to model ne seekha nahi — **ratt liya**. Aur marks jhootha high aa jaayenge.

Isliye humne bataa split **photo ke hisaab se** kiya, tukde ke hisaab se nahi: **338 photos padhai ke liye, 112 photos exam ke liye**. Code me ek check bhi laga diya hai jo galti hone par program crash kar dega.

3. Har Model Kya Use Kar Raha Hai Sawaal 2

Model 1 — Domain Gate (Reception)

Use karta hai: simple image statistics — image kitni grayscale hai, usme radar wala grainy noise (speckle) hai ya nahi, flat area kitna hai, colour kitna hai.

Do hi jawab deta hai: `sar` (aage bhejo) ya `unsupported` (mana kar do).

Model 2 — U-Net (Junior Doctor)

Use karta hai: ek Convolutional Neural Network, jiske **94,429 parameters** hain. Ye bahut chhota hai — comparison ke liye, famous ResNet-50 me 2.5 crore parameters hote hain.

Input me do cheezein deta hai (ye interesting hai):

Kya de rahe hain	Kyu
1. Saaf ki hui photo	"Kahan kaala hai" — tel dhoondhne ka pehla clue
2. Texture map	"Kahan chikna hai, kahan khurdura" — yahi tel aur look-alike ka farak batata hai. Tel lehren ko dabaata hai, isliye tel kaala aur chikna hota hai. Kam-hawa wala look-alike kaala to hota hai par khurdurapan waisa hi rehta hai.

Output: har pixel ko 5 me se ek label — Sea, Oil, Look-alike, Ship, Land.

Model 3 — Decision Layer (Senior Doctor)

Use karta hai: 6 numbers jo training ke baad calibrate hue. Sabse important:

- Pixel tab tel maano jab uska confidence **60% se upar** ho
- Bahut chote dhabbe ignore kar do (wo noise hai)
- Tel ka evidence look-alike ke evidence se **zyada** hona chahiye
- Hawa 3 m/s se kam ya 12 m/s se zyada → confident call mat do

5 jawab de sakta hai:

Jawab	Matlab
SPILL	Tel hai
REVIEW	Pakka nahi bol sakta — insaan dekhe. Ye galat jawab nahi hai
LOOKALIKE	Kaala dhabba hai par tel nahi
CLEAN	Saaf hai

Model 4 — Drift (Physics Wala Banda)

Use karta hai: ek simple sa equation aur uska careful integration:

Tel ki speed = paani ki speed + (hawa ki speed ka 3%)

Us 3% ko "**3% rule**" kehte hain — ye NOAA (America ki agency) ke official oil spill software me bhi yahi hai. Saath me Coriolis effect bhi lagta hai (prithvi ghoomti hai isliye hawa wala hissa thoda daaye mud jaata hai).

Model 5 — Data Generator

Use karta hai: physics formulas se nakli SAR photo banata hai — samundar ka speckle noise, tel ka dark smooth patch, jahaz ka bright point, zameen ka textured area.

4. Aur Wahi Kyu Use Kar Rahe Hain Sawaal 3

Domain Gate kyu? — Ek asli galti se bachne ke liye

U-Net sirf **radar** photos par train hua hai, jahan tel **kaala** hota hai. Lekin normal camera/drone ki photo me tel **chamakdar silver ya rainbow** dikhta hai — bilkul ulta!

Agar aisi photo bina check ke chali jaye:

- Asli spill ki drone photo → model kaala dhabba dhoondhega, chamakdar tel miss karega → **"saaf hai" bol dega**
- Normal samundar ki photo → koi bhi kaala hissa mil jayega → **"tel hai" cheekh dega**

Dono hi bahut kharab hain. Isliye pehle check, phir analysis.

U-Net kyu, koi aur network kyu nahi?

Kyunki hume sirf ye nahi janna ki "photo me tel hai ya nahi" — hume janna hai **tel exactly kahan hai**, uska aakar kya hai, kitna bada hai. Isko **segmentation** kehte hain, aur U-Net iske liye hi bana tha (originally medical images ke liye).

Model itna chhota kyu rakha (sirf 94,429 parameters)?

- Laptop ke CPU par minute me train ho jaata hai — GPU ki zaroorat nahi
- Hamare paas data kam hai. Bada model kam data par **ratt** leta hai (overfitting)
- Demo ke waqt kisi bhi machine par chal jaata hai

Drift ke liye AI kyu nahi?

Kyunki tel ka behna **exactly calculate** ho sakta hai. Aur iska ek bahut bada fayda hai — ise **verify** kiya ja sakta hai.

Humne isko **10 aise cases** ke against check kiya jinka jawab maths se pehle se pata hai. Ek known point se tel chhoda, aur model ne wo point **200 metre ke andar** wapas dhoondh liya.

AI model ke saath ye kabhi nahi kar sakte — wo bas "achha lag raha hai" tak hi rahega.

Do baar U-Net kyu likha (NumPy aur PyTorch dono)?

Version

Kyu

NumPy wala	Bina kuch install kiye chalta hai. Aur ye dikhata hai ki hume maths aati hai — sirf library nahi chala rahe.
PyTorch wala	Training me 10 guna fast. Presentation me jo tech stack likha hai wahi.

Dono bilkul same network hain — humne test karke check kiya: probability maps **0.0000015** tak match karte hain, aur masks **100% same** aate hain.

Accuracy ke bajaye precision/recall kyu?

Samundar ka **90% hissa saaf** hota hai. To ek aisa model jo hamesha "koi spill nahi" bole, wo bhi **90% accuracy** le aayega — aur bilkul bekaar hoga.

Isliye hum accuracy quote hi nahi karte. Hum ye batate hain: **precision** (jab "tel hai" bola to kitni baar sach tha) aur **recall** (asli tel me se kitne pakde).

5. Tech Stack — Kya Kya Laga Hai Sawaal 6

Backend (server side)

Language: Python

Cheez	Kaam kya hai	Kyu chuna
Starlette / FastAPI	Web framework — API banata hai	Starlette FastAPI ka hi andar ka engine hai. FastAPI installed ho to wo chalta hai (bonus me auto documentation /docs mil jaata hai), warna Starlette.
Uvicorn	Server jo chalata hai	Fast, standard
NumPy	Poora neural network isi par	Bina install ke chal jaata hai
SciPy	Filters, shapes dhoondhna	Image processing ke tools
OpenCV	Resize, boundary nikalna	Tel ke kinaare ka outline
Pillow	Photo padhna/likhna	PNG, JPEG support
PyJWT	Login token	Secure session
SQLite	Database	Python me built-in — zero setup
PyTorch (optional)	Training	10 guna fast training
rasterio (optional)	GeoTIFF padhna	Asli satellite file me coordinates hote hain → asli km ² aur asli map location
PostgreSQL + PostGIS (optional)	Bada database	Production ke liye. Code already ready hai — sirf ek setting badalni hai

Frontend (jo browser me dikhta hai)

Frontend me koi framework NAHI hai

Na React, na Angular, na Vue. Na Tailwind, na Bootstrap. Na `npm install`, na build step, na `node_modules` folder.

Sirf **plain HTML + CSS + JavaScript**.

Kyu? Kyunki poora project ek folder hai. `python run.py` chalao, chal gaya. Demo ke din jab time kam ho aur internet slow ho, ye bahut bada fayda hai.

Language / Tool	Lines	Kaam
HTML	471	Do page — landing/login, aur main console
CSS	1,728	Poora design — dark theme + light theme, saare colours, animations, mobile responsive. Sab khud likha.
JavaScript	1,295	Saara interaction — upload, results, routing, map, charts
Leaflet	library	Interactive map jisme tel ka area dikhta hai
Canvas / SVG	—	Landing page ki lehrein, drift ka chart

Frontend ke kuch smart decisions

- **Hash routing** — URL me `#detect` , `#reports` . Page reload nahi hota, koi router library nahi chahiye
- **Theme pehle apply hoti hai** — page dikhne se pehle hi, warna ek second ke liye safed flash aata
- **Cache busting** — CSS/JS ke naam me version number, warna browser purani file dikhata rehta hai (ye problem humein actually aayi thi)
- **100% responsive** — 5 alag screen size par test kiya. 880px se choti screen par sidebar slide-out drawer ban jaata hai
- **LIVE vs DEMO labels** — jo panel asli data dikhata hai uspe "live" likha hai, jo placeholder hai uspe "DEMO". Dono kabhi mix nahi hote, taki koi nakli data ko asli na samajh le

6. Kaunsi Tech Kahan Lagi + Poora Flow Sawaal 7

Jab aap ek photo upload karte ho, andar kya hota hai

- 1 **Browser me file drop** — JavaScript file uthata hai, saath me scene size aur wind speed (agar di ho) jodta hai.

Tech: JavaScript · File: `js/app.js`

- 2 **Server ko bheja** — login token ke saath.

Tech: fetch API · File: `js/api.js`

- 3 **Server file receive karta hai** aur check karta hai — 25 MB se badi to nahi, format sahi hai, wind value 0-60 ke beech hai.

Tech: Starlette · File: `backend/app.py` , `services.py`

- 4 **Reception check (Domain Gate)** — SAR photo hai ya nahi? Nahi to **yahin ruk jao**, "NOT ANALYSED" bhej do.

File: `ml/domain.py`

- 5 **Lab technician (Preprocessing)** — photo saaf hoti hai: contrast theek karo → texture map banao → normalise karo → speckle noise hatao (Lee filter) → 128×128 me resize karo.

Tech: NumPy, SciPy, OpenCV · File: `ml/preprocess.py`

- 6 **Junior doctor (U-Net)** — har pixel ko 5 me se ek label milta hai, confidence ke saath.

Tech: PyTorch (ya NumPy) · File: `ml/predict.py` , `unet.py`

- 7 **Senior doctor (Decision)** — thresholds lagao, chote dhabbe hatao, hawa check karo → final verdict + **reasons**.

File: `ml/decide.py`

- 8 **Result taiyaar karo** — tel ke area ka outline nikalo, use map ke format (GeoJSON) me badlo, aur ek coloured overlay picture banao.

Tech: OpenCV · File: `ml/postprocess.py`

- 9 **Database me save** — aapke account ke saath.

Tech: SQLite · File: `backend/database.py`

- 10 **Browser me dikhao** — result card, overlay image, aur map par tel ka area.

Tech: JavaScript + Leaflet · File: `js/app.js` , `ops.js`

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(Optional) Drift chalao — "ye tel 12 ghante pehle kahan tha?"File: `ml/drift.py`**Har verdict ke saath reason bhi aata hai**

```
"oil covers 0.88% of the scene (2.9x the threshold)
and outweighs look-alike evidence by 3.50x"
```

```
"wind speed not supplied - cannot rule out a low-wind look-alike"
```

Yani system sirf jawab nahi deta — **apna kaam dikhata hai**. Ye investigation system ke liye zaroori hai. Court me evidence chahiye hota hai, sirf ek score nahi.

Login aur privacy**Password**

Kabhi save nahi hota. Sirf uska scrambled hash (PBKDF2, 2 lakh rounds)

Session

JWT token, 8 ghante baad khud expire

Privacy

Har detection sirf uske owner ko dikhti hai. Account page se sab delete kar sakte ho

7. Aage Kya Kar Sakte Hain Sawaal 4

Kya karein	Kya milega	Mehnat
Zenodo Part I download karein 1,200 tel wali photos, 40 GB	Training data 8 guna. Accuracy achhi khaasi badhegi	Sirf download + 2 command
Jahaz ki pehchaan (AIS)	Pata chalega kis jahaz ne tel giraya — PPT ka wada poora hoga	Bada kaam — naya module, naya screen
Live satellite feed	Manual upload khatam — automatic 24×7 monitoring	Bada kaam
GPU par training	45 min ka kaam 3 min me	Kuch nahi — code khud detect karta hai
PostgreSQL par shift	"Is area me pichle mahine kitne spill hue" jaise sawaal pooch paoge	Sirf ek setting
Optical photo bhi milao	Do source se confirmation → galat alarm kam	Poora retrain karna padega

8. Code Me Kya Change Karne Se Kya Badlega Sawaal 5

Change	Asar	Kaunsi file badlegi
Zyada data se retrain	Accuracy upar	Koi code change nahi — bas 2 command
Model bada karna 94k → 375k parameters	Zyada capacity, par overfit ka risk	Sirf ek flag: <code>--base 16</code>
GPU use karna	15 guna fast	Kuch nahi — automatic
PostgreSQL + PostGIS	Spatial queries	Sirf <code>DATABASE_URL</code> setting — code ready hai
Thresholds badalna	Zyada alerts ya kam alerts	<code>models/decision.json</code> — sirf numbers
Naya class add karna jaise "plastic waste"	6 class ho jayengi	<code>unet.py</code> , <code>datagen.py</code> , naye labels, poora retrain

AIS attribution	Jahaz ki pehchaan	Naya module + naya API + naya screen + database table
Optical fusion	Do source se confirmation	Input 2 channel se 3 — poora retrain

9. Result — Aur Kya NAHI Bolna

84.9%

Precision

"tel hai" bola to sach tha

93.0%

Recall

asli tel me se pakde

0.581

Oil IoU

shape kitna sahi

112 aisi satellite photos par jo model ne training me kabhi dekhi hi nahi thi

Aasan bhasha me:

- System ne 100 baar "tel hai" bola → **85 baar sach tha**, 15 baar false alarm
- 100 asli tel ki ghatnaon me se → **93 pakad liye**, 7 miss ho gaye

100% kyu nahi?

Kyunki radar me tel aur look-alike **sach me** milte-julte dikhte hain — ye physics ki limit hai, coding ki kami nahi.

Duniya ka best published result bhi isi aas-paas hai (Oil IoU 0.538 — hamara 0.581 usse thoda behtar hai).

Aur Europe ka **CleanSeaNet**, jo asli operational system hai aur government use karti hai — uske sirf **80% alerts** hi confirm hote hain.

Isliye hamara system confuse hone par **"REVIEW"** bolta hai — galat jawab dene ke bajaye maan leta hai ki pakka nahi pata.

Ye numbers mat bolna

- **Look-alike, Ship, Land ka score** — is dataset me inke labels hain hi nahi. Inka score 0 aayega aur wo galat samjha jayega.
- **Synthetic (nakli data) ke numbers** — wo 92.6% aata hai, par wo nakli photos par hai. Asli SAR par nahi.
- **"Humne paper ko beat kiya"** — ye mat bolna. Hamara split alag hai. Bolna: *"on our held-out split"*.

Abhi kya nahi hai — khud bata dena

Kya nahi hai	Kya bolna
Jahaz ki pehchaan (AIS)	"Detection aur drift ban chuke hain aur measure ho chuke hain. Attribution design ho chuka hai, abhi bana nahi."
Optical photo fusion	"Abhi sirf SAR chalta hai. Dashboard par optical panel clearly 'DEMO / not ingested' marked hai."
Live satellite feed	"Abhi manual upload hai."
Ek purani file <code>classical.py</code>	"Ye alternative detector humne likha tha par final pipeline me U-Net use ho raha hai."

10. Judge Ke Sawaal — Ready Jawab

Sawaal	Jawab
"Accuracy kitni hai?"	"Accuracy yahan galat metric hai — samundar ka 90% saaf hota hai, to 'koi spill nahi' bolne wala bekaar model bhi 90% accuracy le aayega. Hum precision 84.9% aur recall 93% batate hain, 112 unseen satellite photos par."
"100% accurate kyu nahi?"	"Radar me tel aur look-alike genuinely overlap karte hain — physics ki limit hai. Europe ka operational CleanSeaNet bhi 80% par hai. Isliye hamara system confuse hone par REVIEW bolta hai."
"Data leak to nahi hua?"	"Split satellite photo ke level par hai, tukde ke level par nahi. Code me automatic check laga hai jo leak hone par crash kar dega."
"Ready-made model use kiya?"	"Nahi. Architecture khud likha, backprop ka maths khud likha aur numerical gradient se verify kiya. Sirf 94,429 parameters hain."
"Look-alike ka score 0 kyu?"	"Kyunki dataset me look-alike ka pixel-level label hai hi nahi — sirf itna likha hai ki 'is photo me look-alike hai', kahan hai ye nahi. Hum andaaze ke labels par train nahi karte. Un photos ko hum hard negative ki tarah use karte hain, jisse false alarm kam hote hain."
"Drift verify kaise kiya?"	"Dataset se nahi — 10 maths ke known solutions se. Known point 200 metre ke andar wapas mil jaata hai. Live demo bhi hai: <code>/api/drift/verify</code> ."
"Wind data kahan se?"	"Abhi user deta hai (optional). Na de to system khud bolta hai 'low-wind look-alike rule out nahi kar sakte' — chhupata nahi. Production me weather feed lagega."

"Scale kaise karoge?"

"Modular hai. Database SQLite se PostgreSQL sirf ek setting se — code already ready hai. CPU se GPU bina code change ke."

11. Chalane Ke Commands

```
cd C:\Users\HP\Desktop\NeelDrik

python run.py                # website chalu → http://127.0.0.1:8000

python tests\test_pipeline.py # 39 checks
python tests\test_parity.py   # NumPy aur PyTorch same hain?
python -m backend.ml.drift     # drift ka physics test

python train.py --torch --epochs 40 # dobara train
python scripts\evaluate.py --wind   # detailed evaluation
```

Sab bhool jao, bas ye yaad rakho

Ek chhota sa AI (94,429 parameters) satellite ki radar photo me tel dhoondhta hai. Physics aur calibrated rules uske faisle ko double-check karte hain. Ek alag maths wala model tel ko wapas uske source tak trace karta hai. **Aur jahan system sure nahi hai, wo saaf bol deta hai ki sure nahi hai — galat jawab dene se accha hai jawab na dena.**

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